



University of
Salford
MANCHESTER



SCHOOL OF
**SCIENCE, ENGINEERING
& ENVIRONMENT**

BEng (Hons) Control and Automation

Automation and Robotics

CRN: 50295

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Lecture Content:

Introduction to Control
in Automation

Industrial Automation

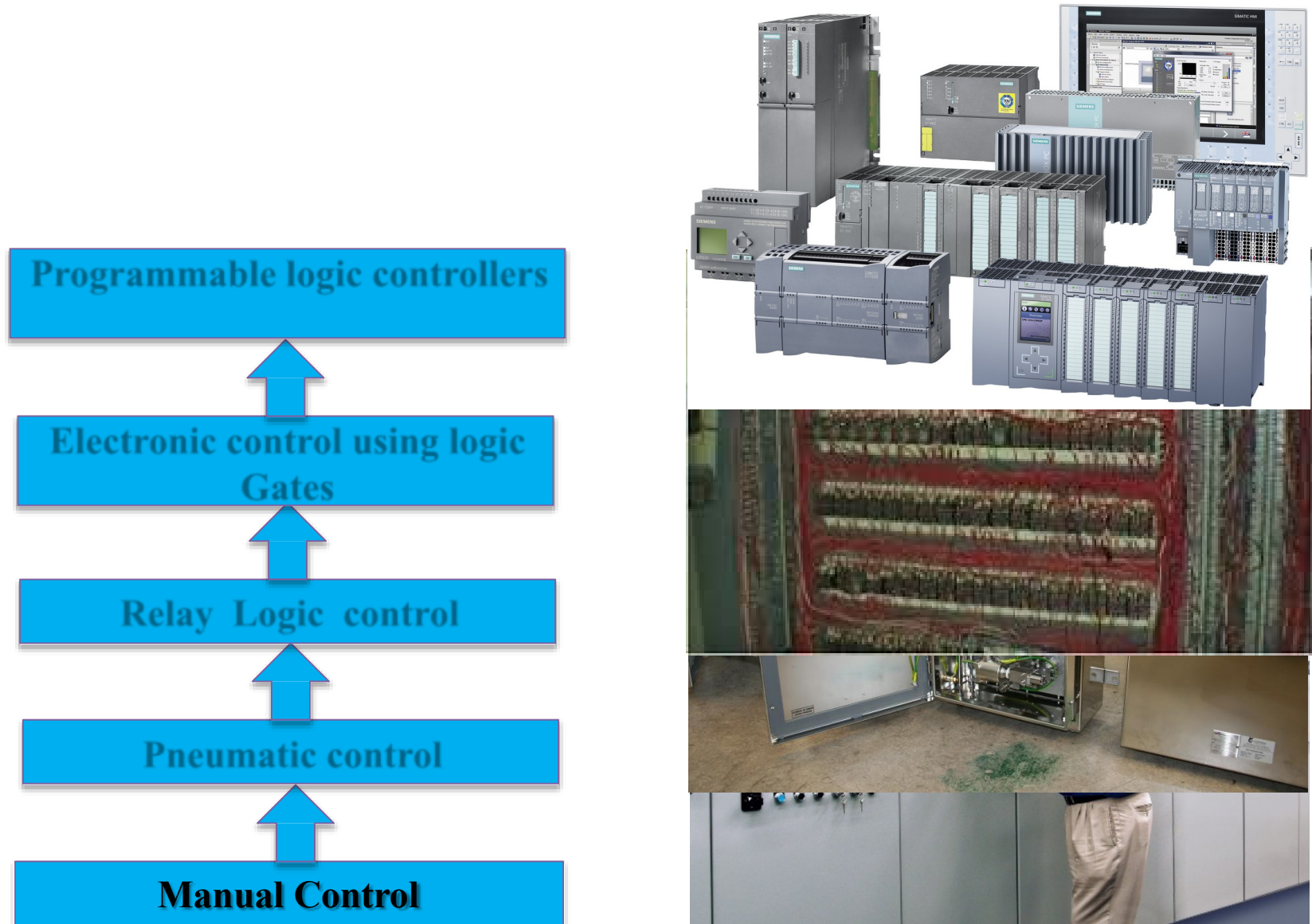
- The technology by which a process or procedure is accomplished without human assistance.
- A technique that can be used to reduce costs and/or to improve quality.
- A process which increases manufacturing speed, while reducing cost.

Which....

- Can lead to products having consistent quality, perhaps even consistently good quality
- Is normally implemented using a program of instructions combined with a control system that executes the instructions.

Quick History

Automation has gone through a number of stages



Industrial Automation

Automation is a technology concerned with application of mechanical, electronic and computer-based system to operate and control system. This technology includes;

- Automatic assembly machines
- Automation machine tools to process parts
- Industrial robots
- Automatic materials handling and storage system
- Automatic inspection system and quality control
- Feedback control and computer process control
- Computer systems for planning, data collection and decision making to support manufacturing activities



Basic Elements of an Automated System

The Actuator

The actuator does the work

- Controlled by the controller.
- The actuator in a automated process may in fact be several actuators, each of which provides an output that drives another in the series of actuator.
- Some actuators can only be on and off. Other actuators respond proportionally with the signal they receive from a controller
- Actuators are ultimately controlled by an electrical signal although this controls other force providing media such as.
 - Electromotive
 - Pneumatic
 - Hydraulic

Basic Elements of an Automated System

The Controller

Tells the actuator to do the work

- A controlled system either may be a digital system or an analogue system
- Digital and analogue controllers are available 'off the shelf' so that systems can be constructed inexpensively and with little specialized knowledge required.
- Modern automated control is almost exclusively PLC based

Basic Elements of an Automated System

The Sensors

provides feedback to the controller so that it knows the actuator is doing the work

- Obviously, controlled automation requires devices to sense system output.
- Sensors also can be used so that a controller can detect and respond to changing conditions in its working environment
- Switches and transducers are another name for sensors.
 - Switches can detect when a measured condition exceeds a pre-set level. Example: closes when a work-piece is close enough to work on.
 - Transducers can describe a measured condition. Example: output increased as level in tank raises.

Automation

- Robots just one tool in automation
- There are different types of automation and equipment used in it
- Selection of the appropriate type depends mainly on product variability and production volume

Automation

- **Fixed (hard) automation**
 - Specialised machines
 - Low variety
 - High volume processes
 - Maximum efficiency
 - High equipment cost but high volume lowers manufacturing cost



Automation

- **Programmable automation**
 - Low production volume
 - High product variety
 - Adaptable to variations but long change over times
 - Control of a 'program'
 - E.g. CNC machine making low volume specialist parts
 - Low volume but premium charged because of this



Automation

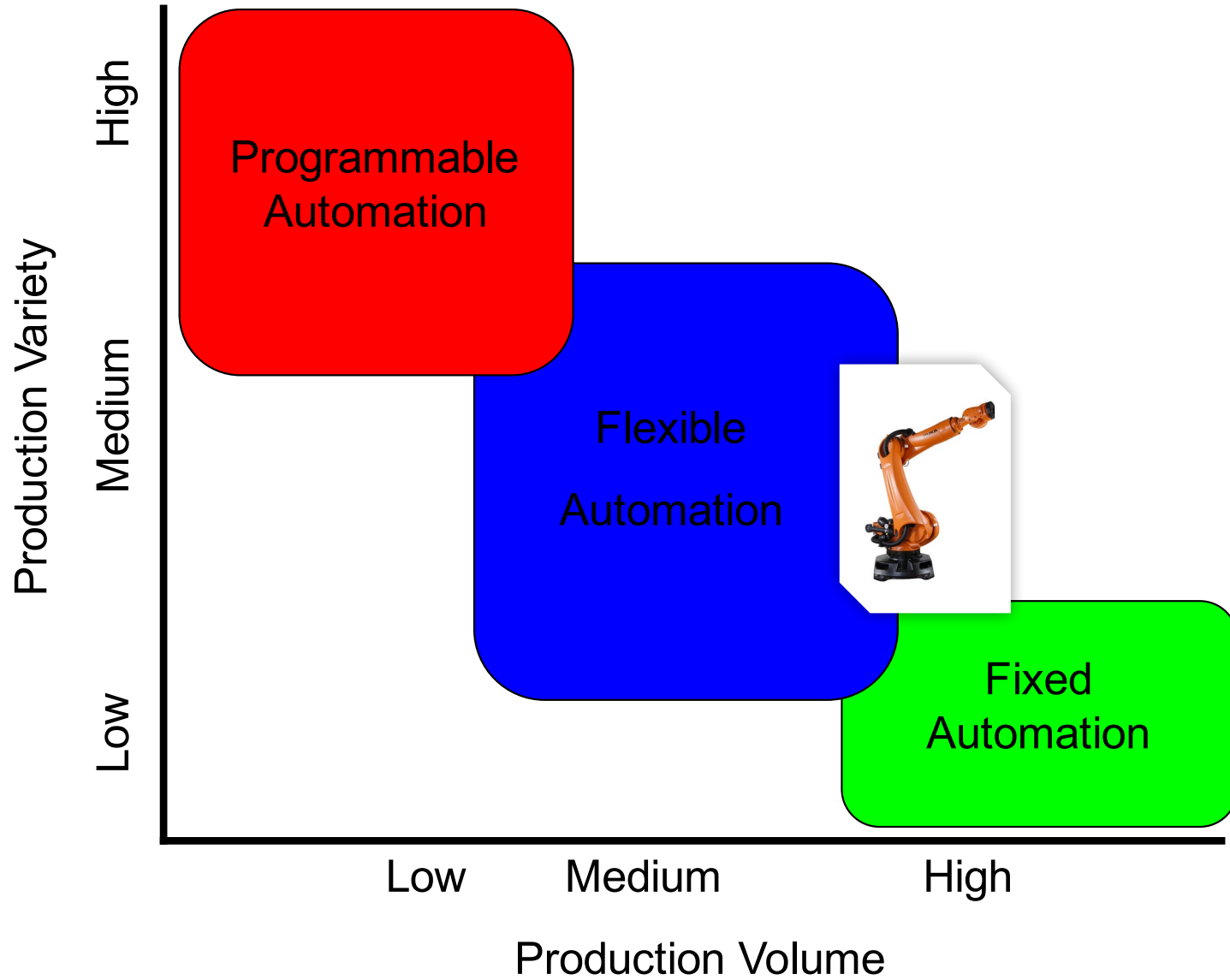
- **Flexible automation**

- Between fixed & programmable automation
- Mid Volumes
- Some product variation and quick change over times
- Also called flexible manufacturing systems (FMS)
- Can be 'programmed'
- Robots considered flexible automation

Automated Machine

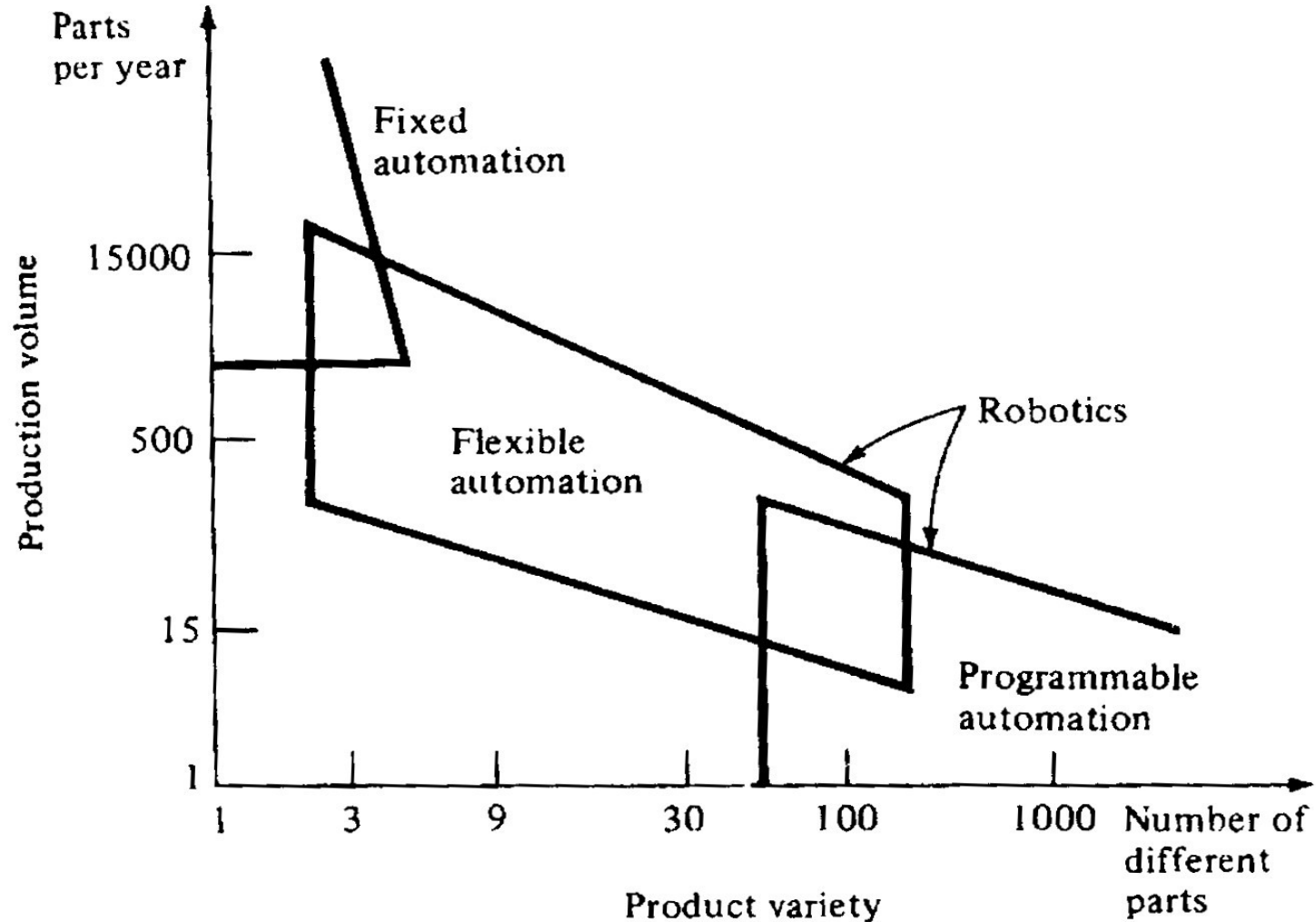
- Examples
 - Fixed Automation
 - Coke putting lids on bottles
 - The production of washer or screws
 - Programmable Automation
 - An F1 team producing an engine block
 - Airbus machining wing spars
 - Flexible Automation
 - Ford making 10 versions of the Focus
 - Ginsters making sandwiches

Automation



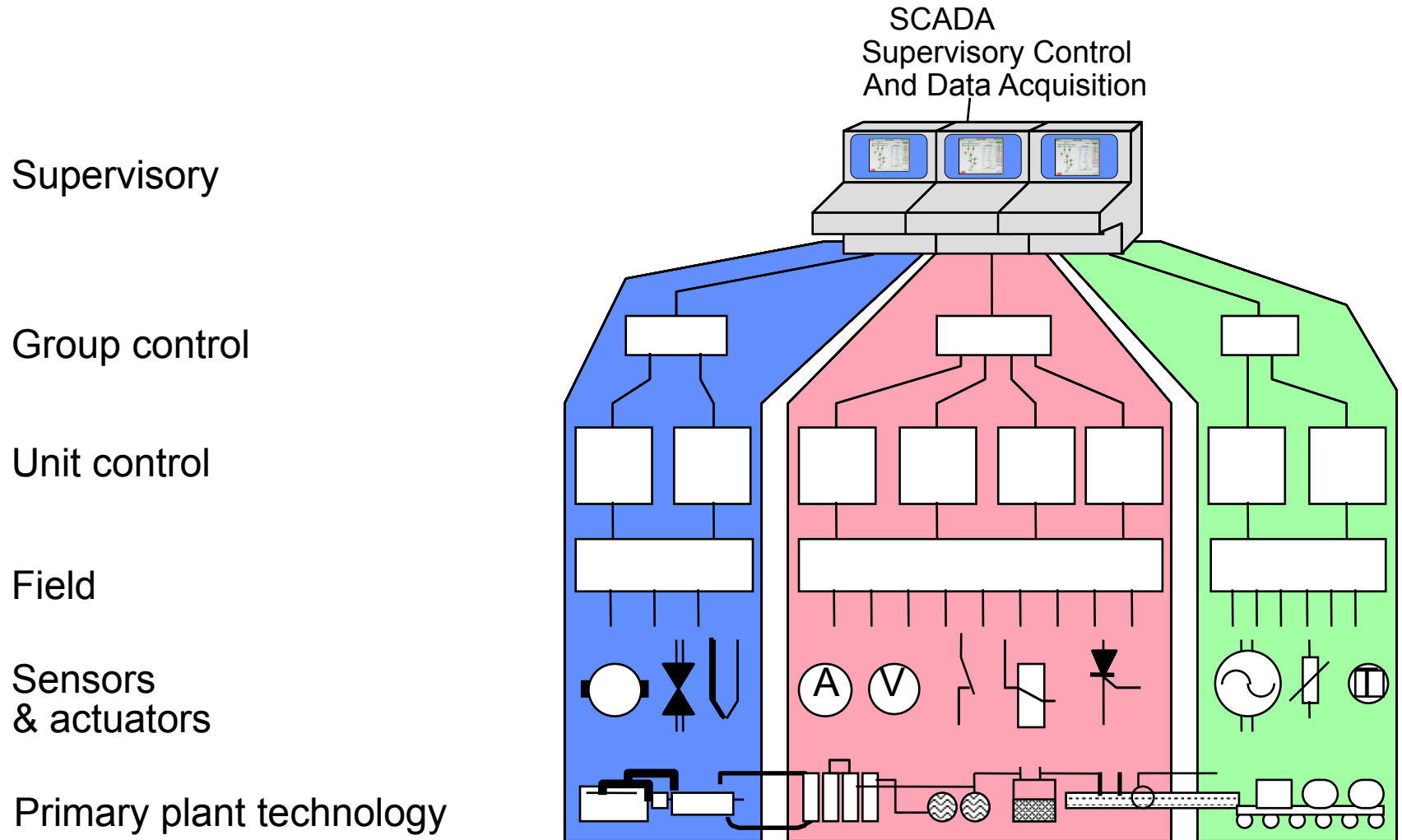
Manufacturing System Classification

Relationship between manufacturing systems



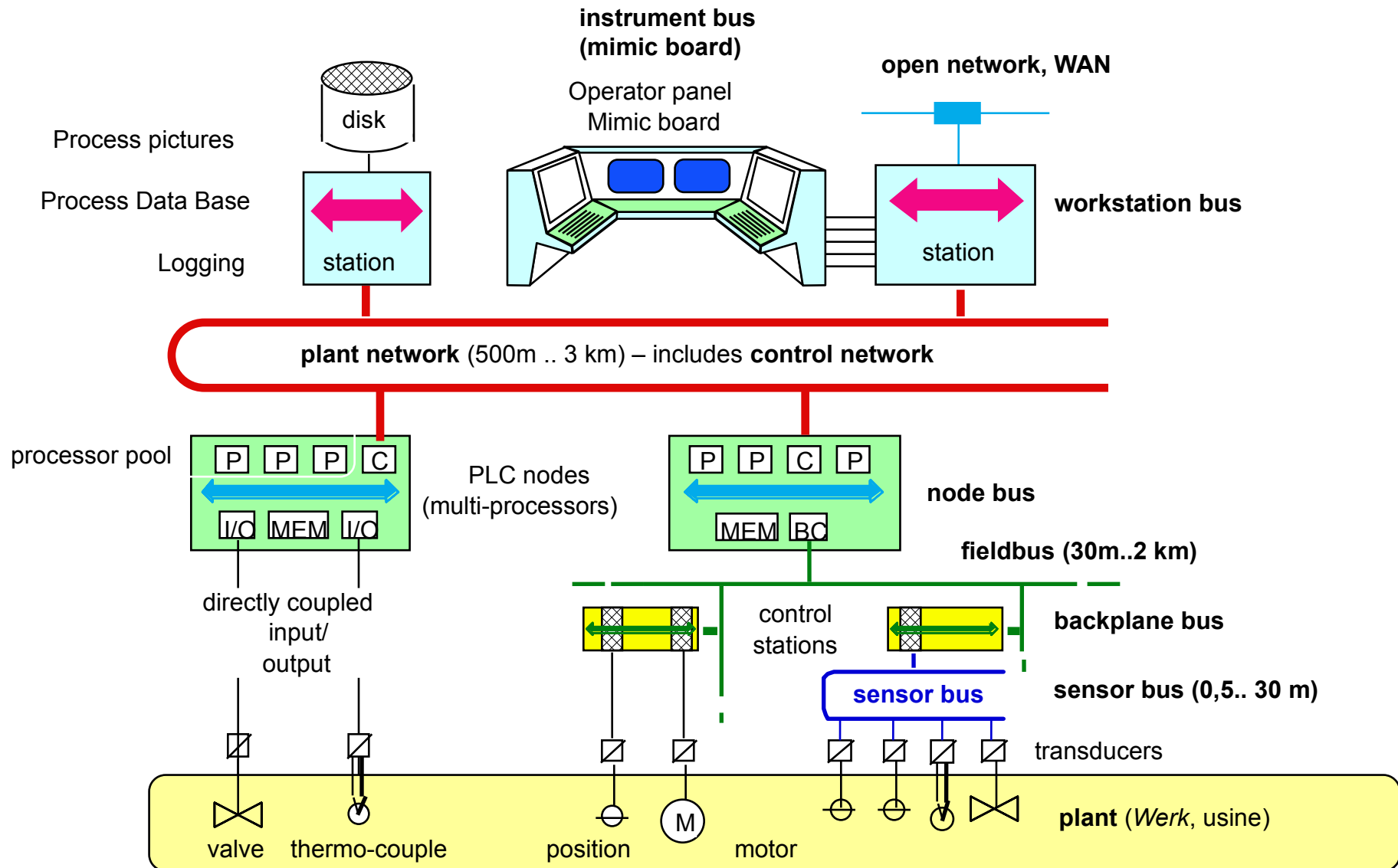
Large Control System Hierarchy

Manufacturing plants can have an extensive hierarchical control system



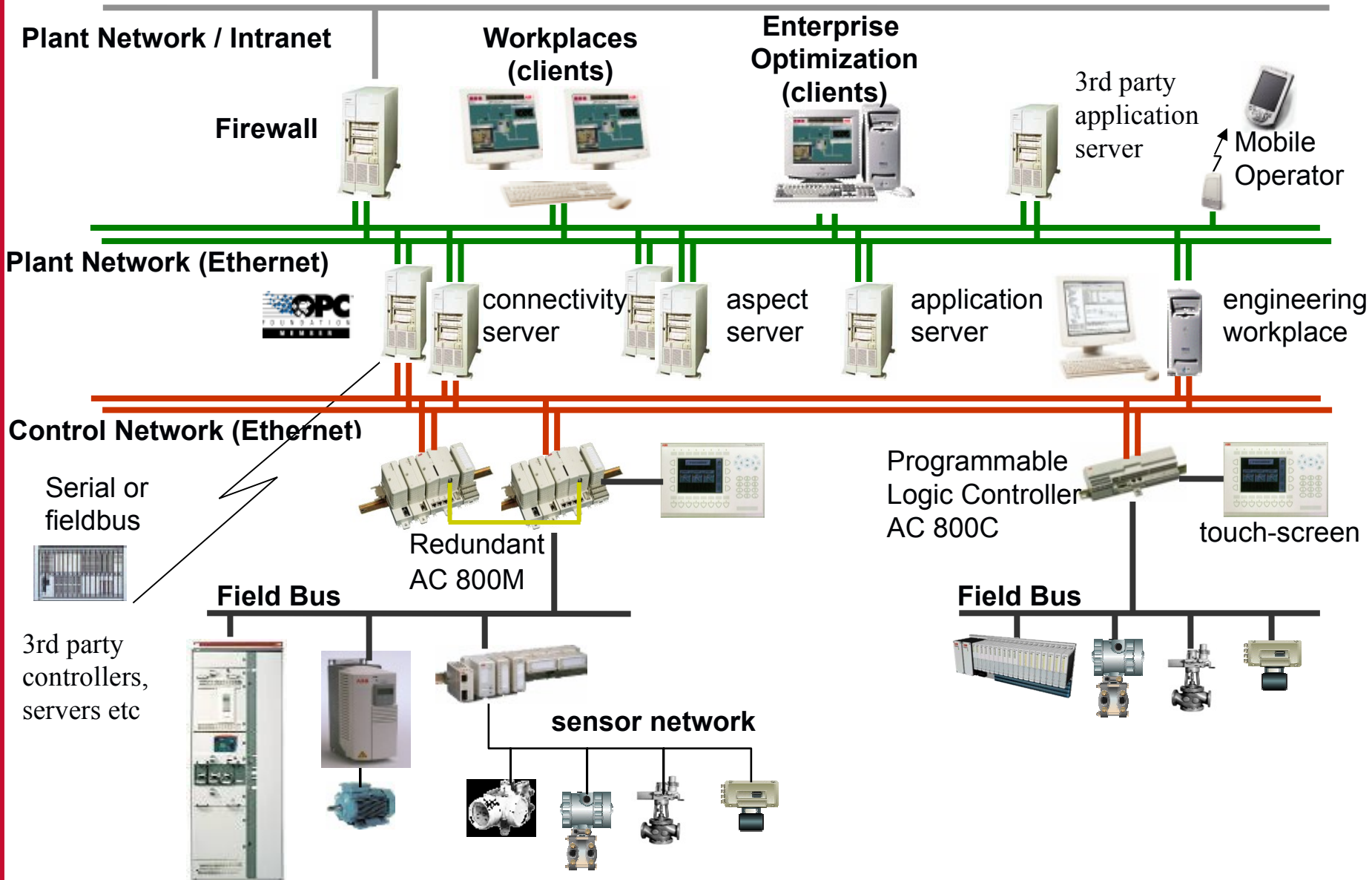
Large Control System Hierarchy

Buses and processors in industrial plants form this hierarchy



Large Control System Hierarchy

ABB Industrial IT (redundant system)



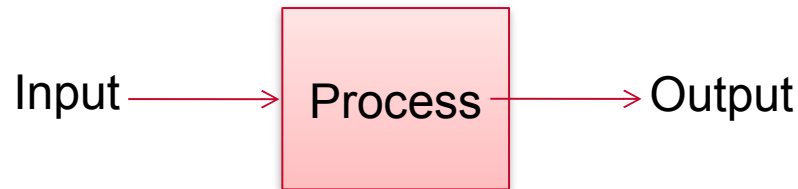
Fundamentals of Control Systems

Definitions

System – An interconnection of elements and devices for a **desired purpose**.

Control System – An interconnection of components forming a system configuration that will provide a desired response.

Process – The device, **plant**, or system under control. The input and output relationship represents the cause-and-effect relationship of the process.



Fundamentals of Control Systems

Definitions

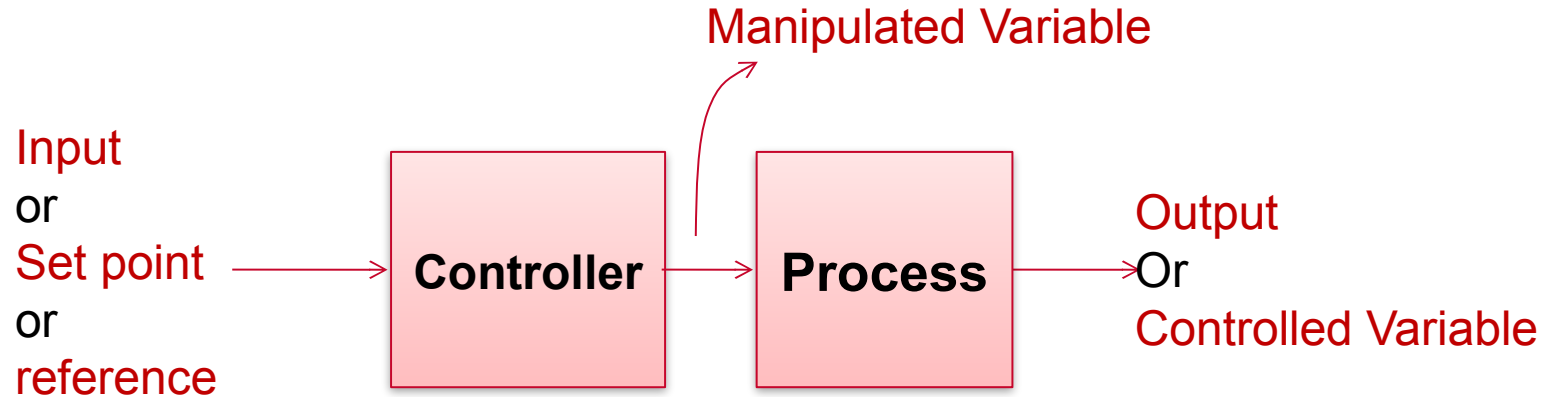
Controlled Variable— It is the quantity or condition that is measured and Controlled. Normally *controlled variable* is the output of the control system.

Manipulated Variable— It is the quantity of the condition that is varied by the controller so as to affect the value of *controlled variable*.

Control – Control means measuring the value of the *controlled variable* of the system and applying the *manipulated variable* to the system to correct or limit the deviation of the measured value from a desired value.

Fundamentals of Control Systems

Definitions

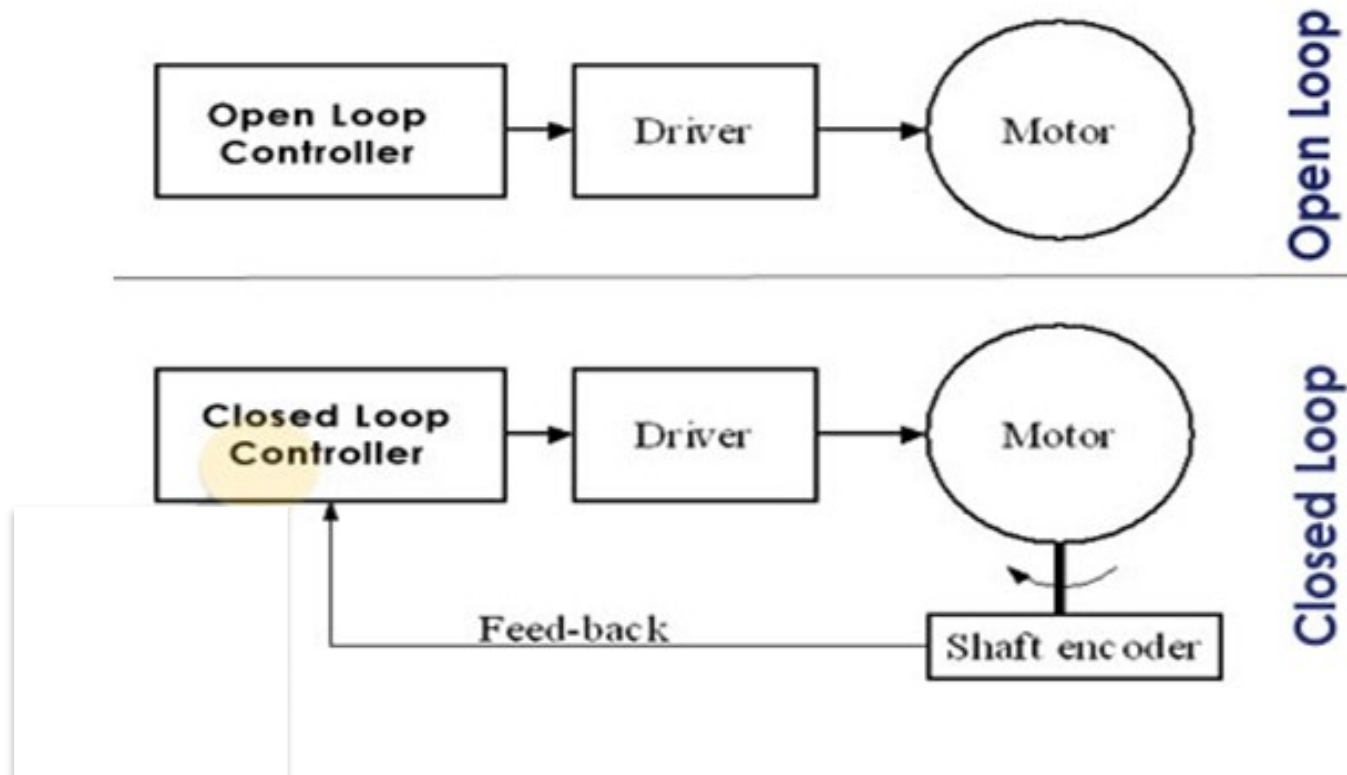


Disturbances— A disturbance is a signal that tends to adversely affect the value of the system. It is an unwanted input of the system.

- If a disturbance is generated within the system, it is called *internal disturbance*. While an *external disturbance* is generated outside the system.

Fundamentals of Control Systems

- If a human operator is available to monitor and control a manufacturing process, **open loop control** may be acceptable

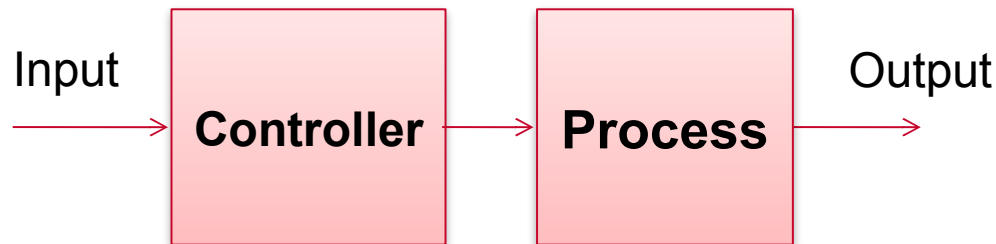


- If a manufacturing process is automated, then it requires **closed loop control**, also known as **feedback control**.

Fundamentals of Control Systems

Open-Loop Control Systems utilize a controller or control actuator to obtain the desired response.

- Output has no effect on the control action.
- In other words output is neither measured nor fed back.

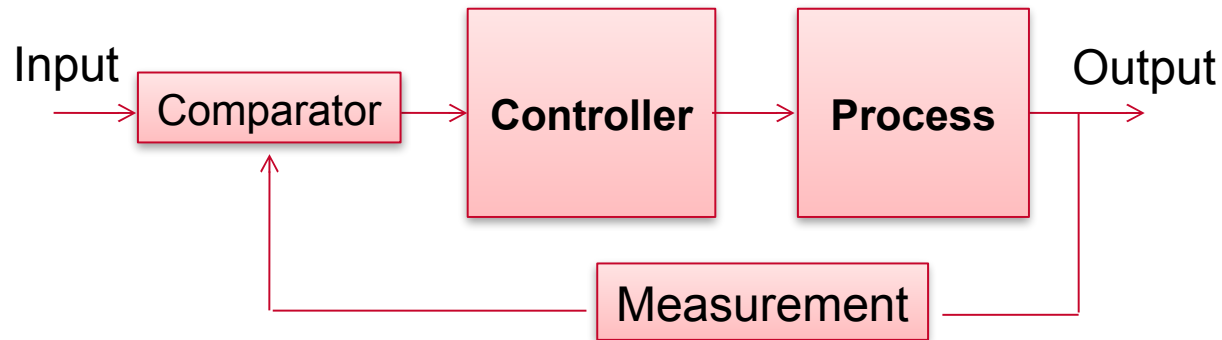


Open-loop control system (without feedback).

- Since in open loop control systems reference input is not compared with measured output, for each reference input there is fixed operating condition. Therefore, the accuracy of the system depends on calibration.
- The performance of open loop system is severely affected by the presence of disturbances, or variation in operating/ environmental conditions.

Fundamentals of Control Systems

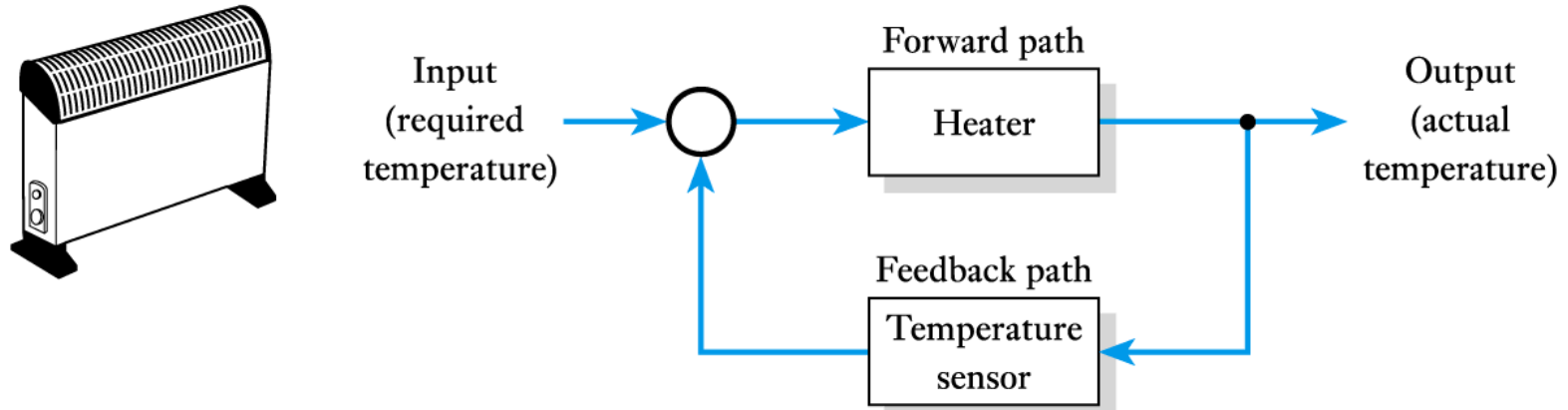
Closed-Loop Control Systems utilizes feedback to compare the actual output to the desired output response.



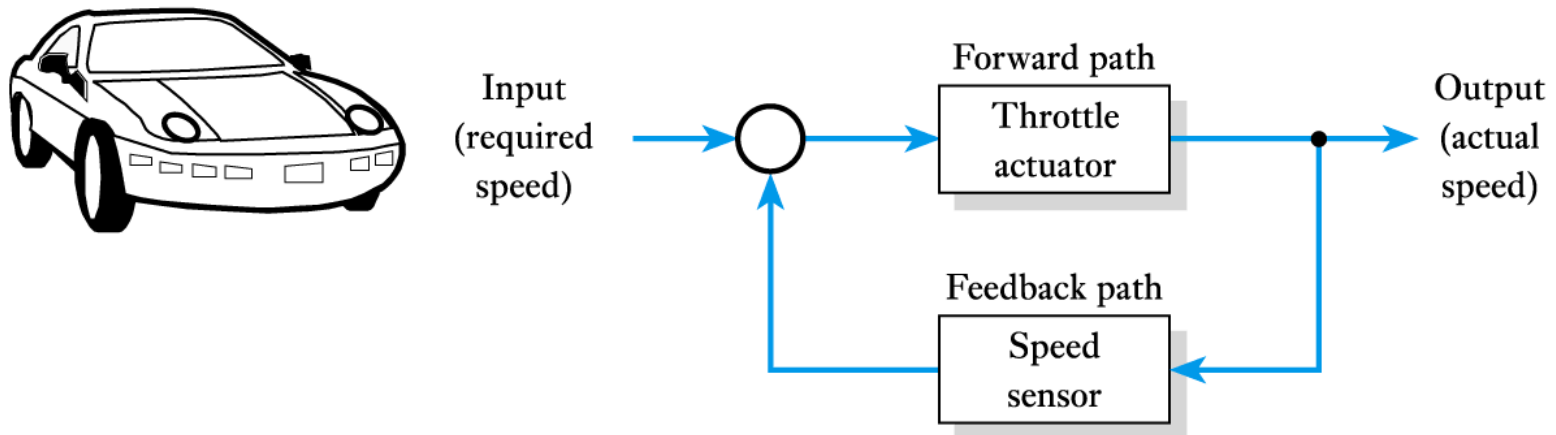
Closed-loop feedback control system (with feedback).

Fundamentals of Control Systems

Temperature control using a room heater

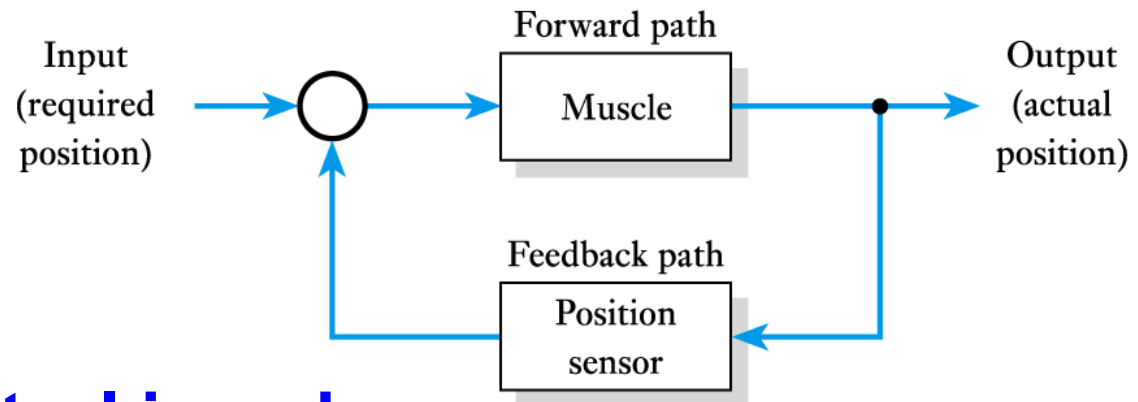


Cruise control in a car

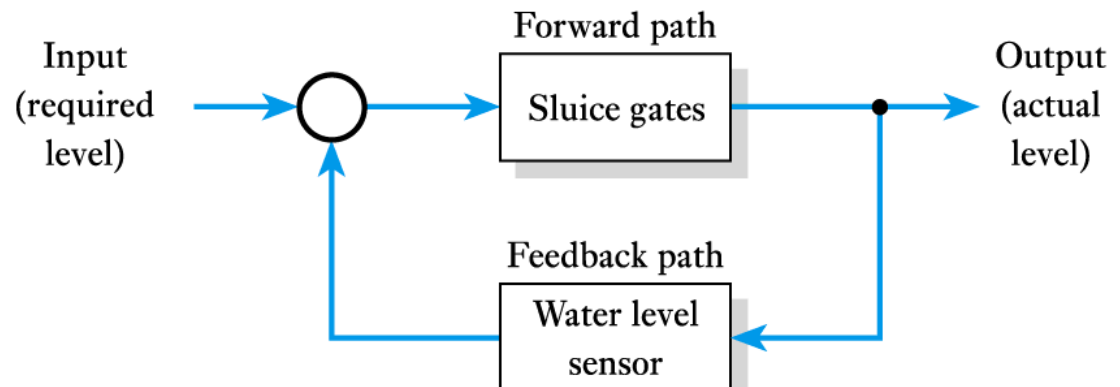
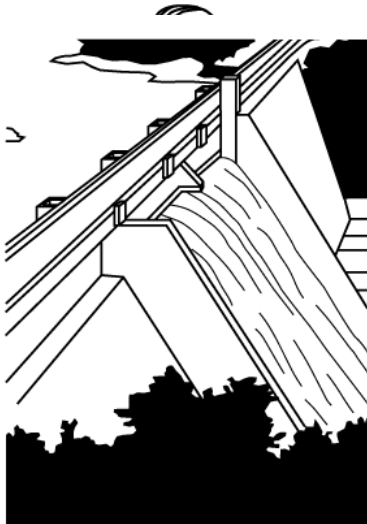


Fundamentals of Control Systems

Position control in a human limb

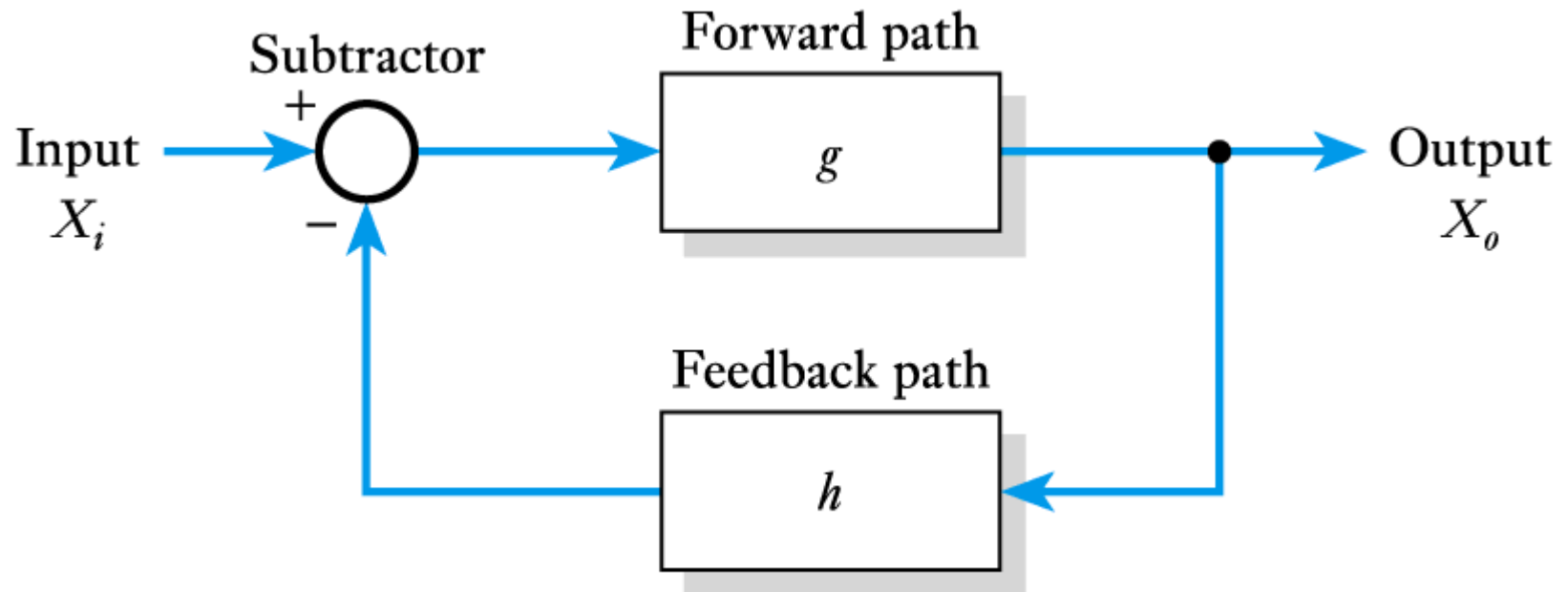


Level control in a dam



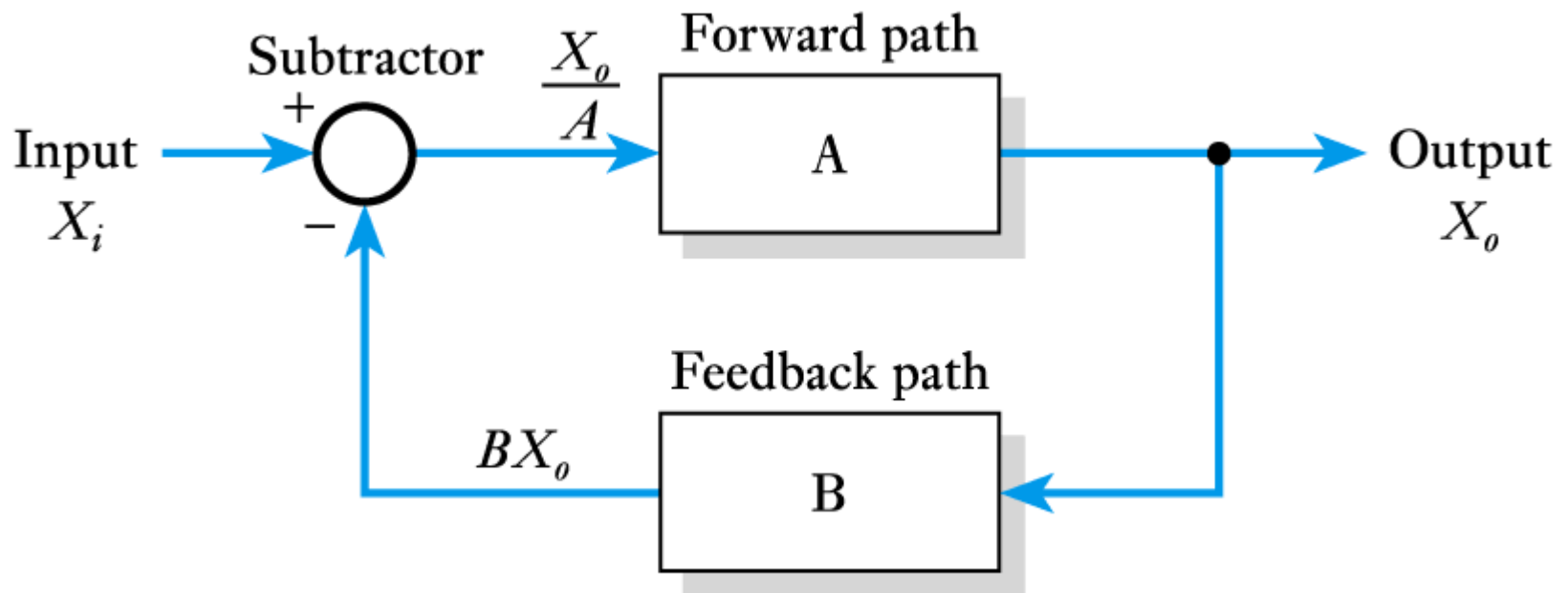
Feedback Systems

A generalised feedback system



Feedback Systems

Adding Values to the Feedback System



$$\frac{X_o}{A} = X_i - BX_o$$

$$\frac{X_o}{X_i} = \frac{A}{1 + AB}$$

Feedback Systems

Thus

$$\text{Overall gain } G = \frac{X_o}{X_i} = \frac{A}{1 + AB}$$

This is the **transfer function** of the arrangement

Terminology:

- A is also known as the **open-loop gain**
- G is the overall or **closed-loop gain**

Feedback Systems

Effects of the product AB

- If AB is negative

- If AB is negative and less than 1, $(1 + AB) < 1$
- In this situation $G > A$ and we have **positive feedback**

- If AB is positive

- If AB is positive then $(1 + AB) > 1$
- In this situation $G < A$ and we have **negative feedback**
- If AB is positive *and* $AB \gg 1$

$$G = \frac{A}{1 + AB} \approx \frac{A}{AB} = \frac{1}{B}$$

- gain is independent of the gain of the forward path A

Feedback Systems

Negative feedback can be applied in many ways

- X_i and X_o could be temperatures, pressures, etc.
- But mainly occurs as voltages and currents

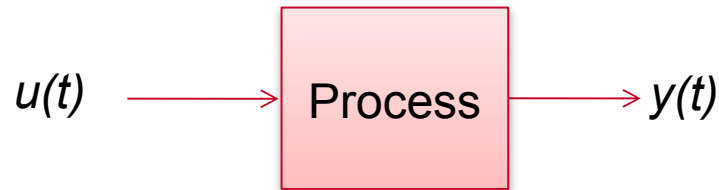
Important in overcoming **variability**

- all active devices suffer from variability
 - their gain and other characteristics vary with temperature and between devices
- using negative feedback we can produce an arrangement where the gain is *independent* of the gain of the forward path
 - this gives us a way of overcoming problems of variability

Types of Control System

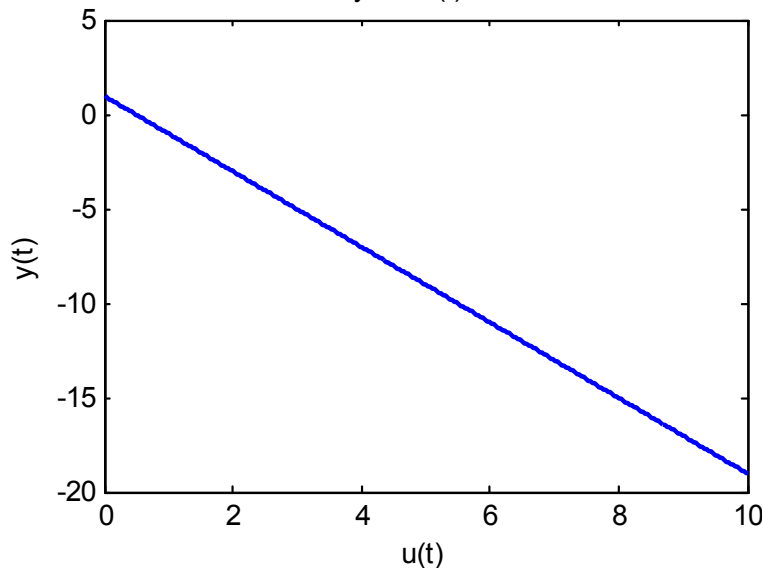
Linear Vs Nonlinear Control System

- A Control System in which output varies linearly with the input is called a linear control system.



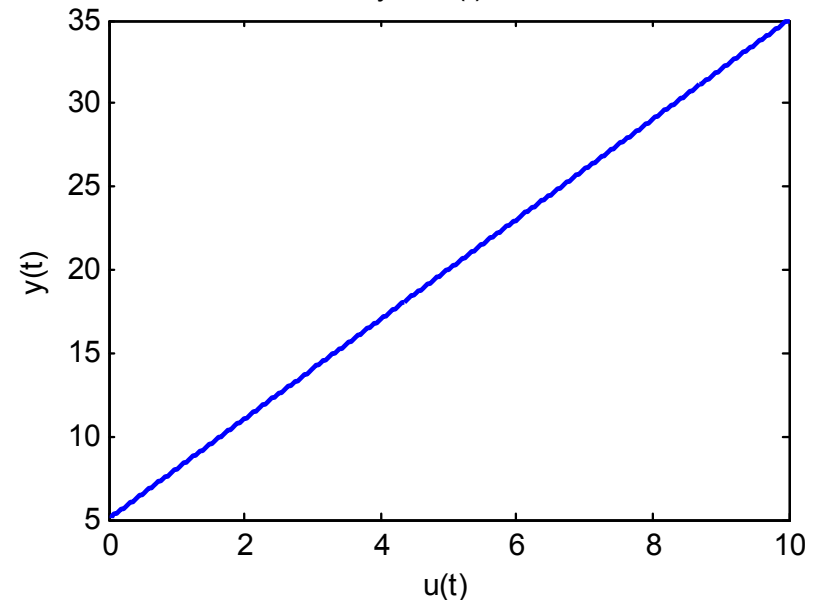
$$y(t) = -2u(t) + 1$$

$$y = -2 \cdot u(t) + 1$$



$$y(t) = 3u(t) + 5$$

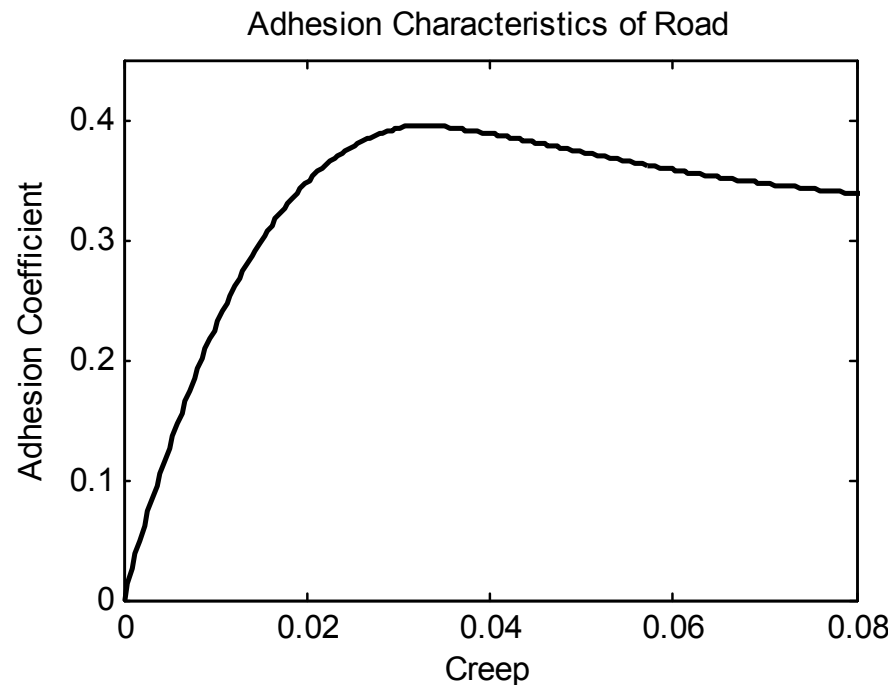
$$y = 3 \cdot u(t) + 5$$



Types of Control System

Linear Vs Nonlinear Control System

- When the input and output has nonlinear relationship the system is said to be nonlinear.



Types of Control System

Time invariant vs Time variant

- When the characteristics of the system do not depend upon time itself then the system is said to time invariant control system.

$$y(t) = -2u(t) + 1$$

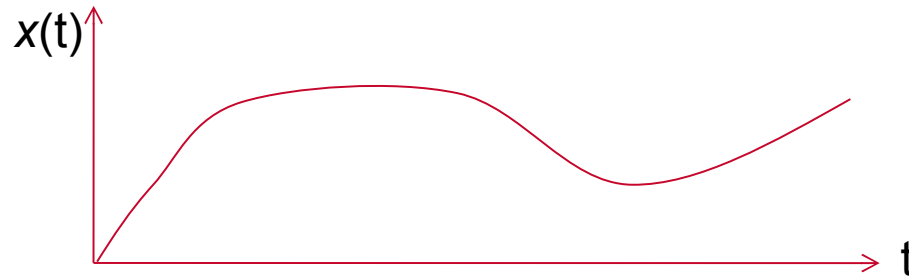
- Time varying control system is a system in which one or more parameters vary with time.

$$y(t) = 2u(t) - 3t$$

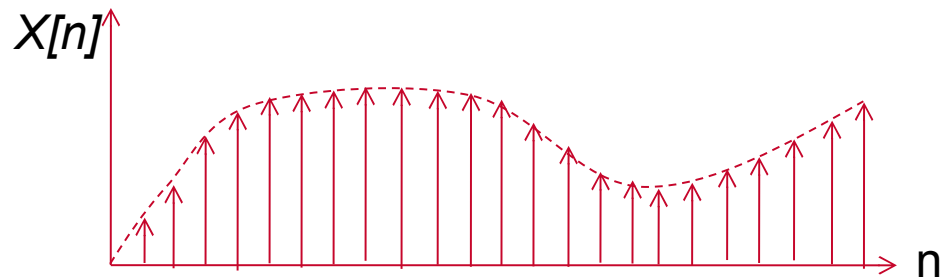
Types of Control System

Continuous Data Vs Discrete Data System

- In continuous data control system all system variables are function of a continuous time t .



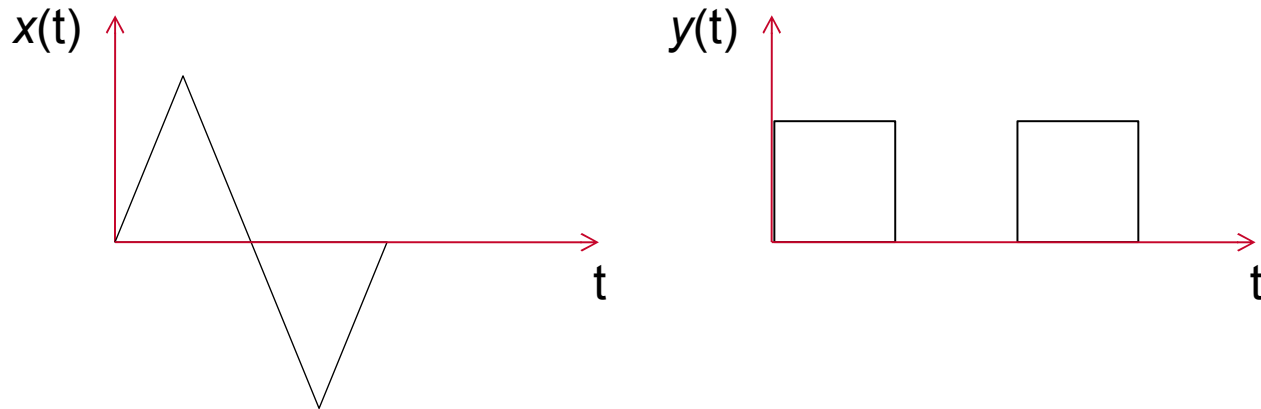
- A discrete time control system involves one or more variables that are known only at discrete time intervals.



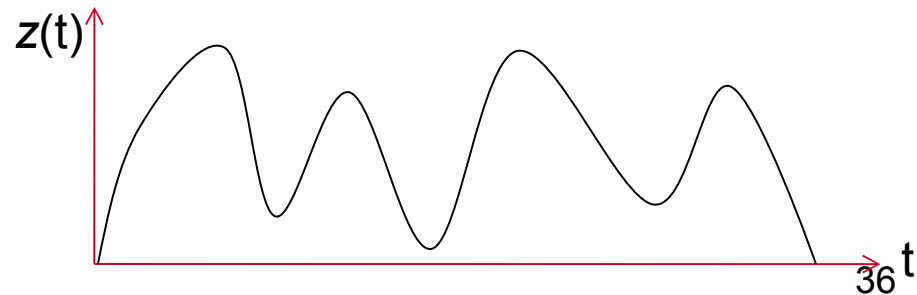
Types of Control System

Deterministic vs Stochastic Control System

- A control System is deterministic if the response to input is predictable and repeatable

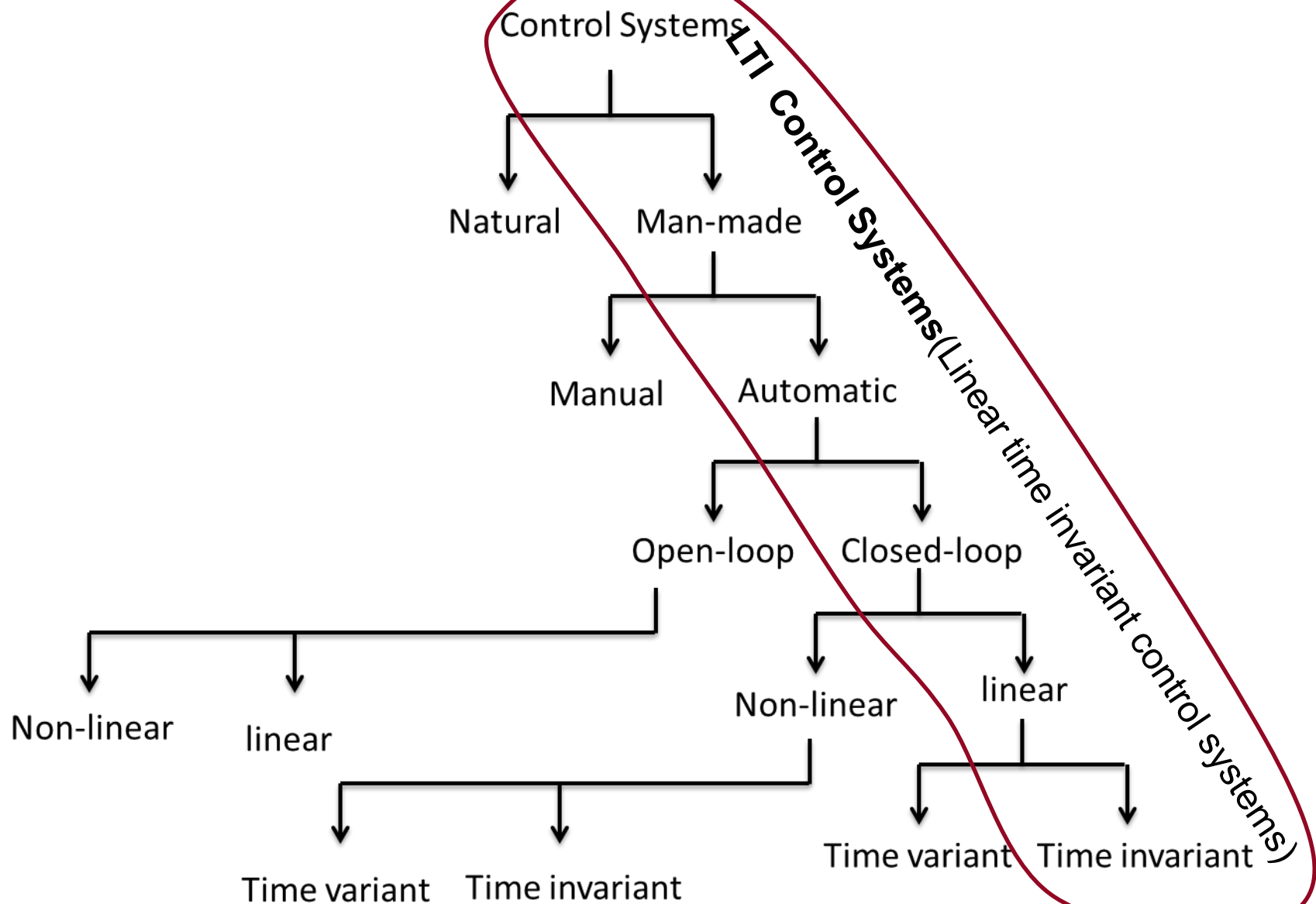


- If not, the control system is a stochastic control system



Types of Control System

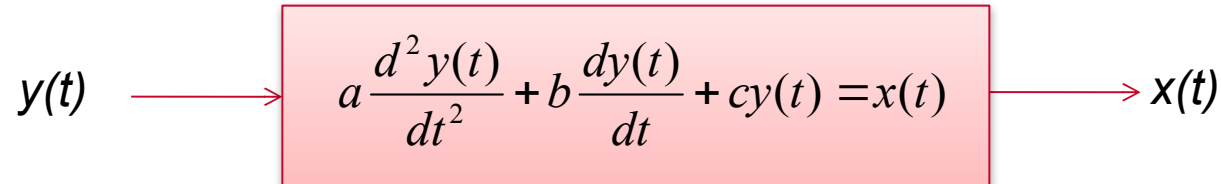
Classification of Control Systems



Transfer Functions

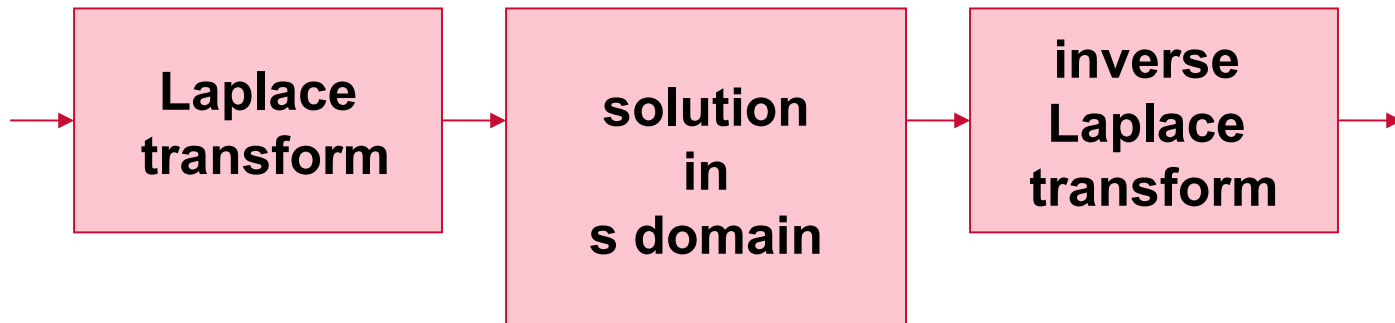
Calculation of control transfer functions

Transfer functions may require the solution of first, second or even higher differential equations



A block diagram showing a differential equation. On the left, the input $y(t)$ has an arrow pointing into a pink rectangular box. Inside the box is the equation $a \frac{d^2 y(t)}{dt^2} + b \frac{dy(t)}{dt} + cy(t) = x(t)$. An arrow points out of the box to the right, leading to the output $x(t)$.

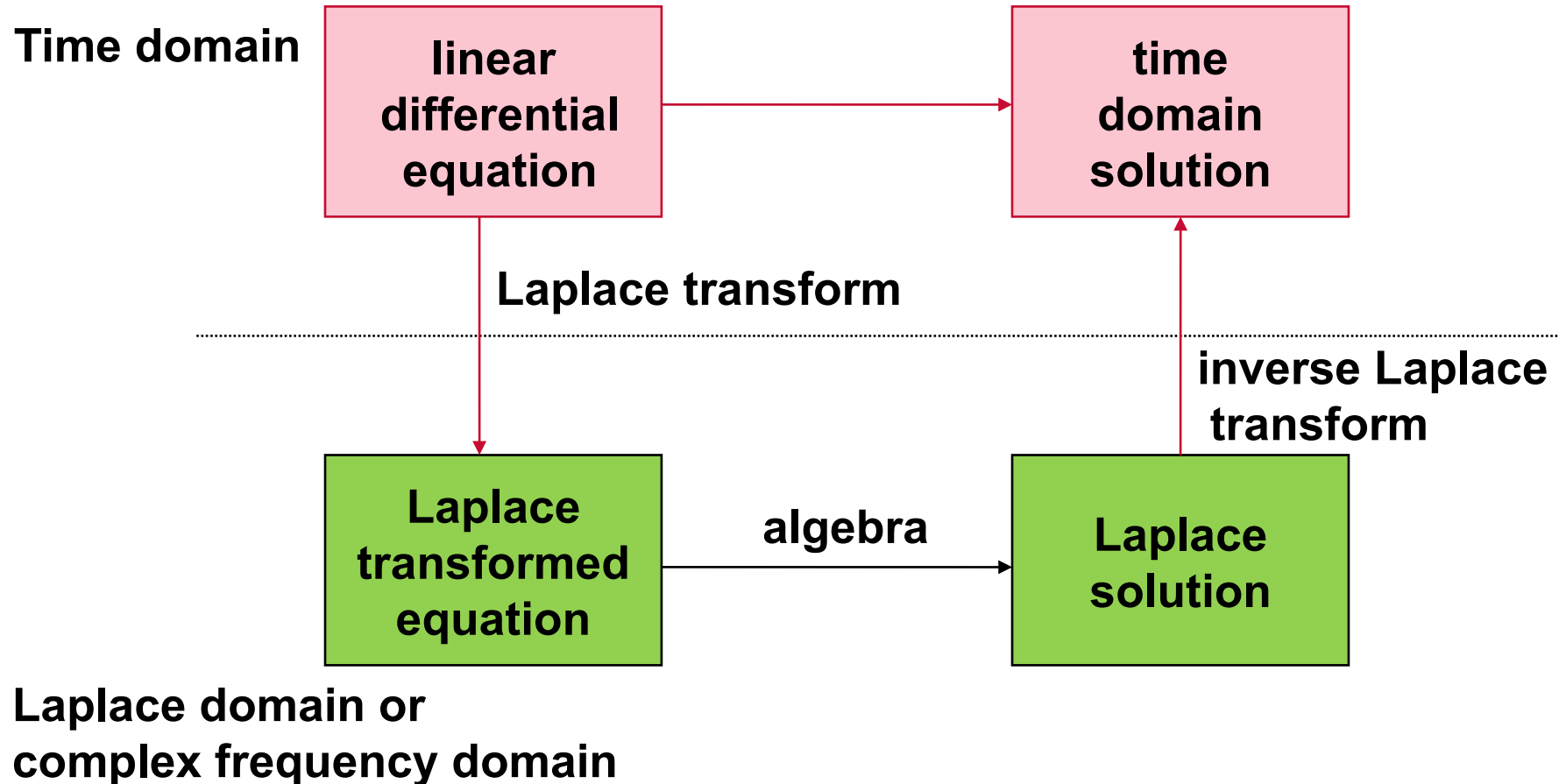
To make this problem easier to solve we use Laplace Transforms



Transfer Functions

Laplace Transform Domain

Transfer functions may require the solution of first, second or even higher differential equations



FOR TOMORROW

Have a look at the two YouTube videos

Laplace Transforms

<https://www.youtube.com/watch?v=ZGPtPkTft8g>

Transfer Functions

<https://www.youtube.com/watch?v=RJleGwXorUk>

Links available on blackboard